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Correlation of King-Devick Test and Helmet Impact Exposures Over a Youth Football Season

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Background: The cumulative effect of repetitive subconcussive head impacts on neurocognitive function during youth contact sports remains largely unknown. There is a paucity of literature evaluating cumulative helmet forces over a season and their correlation with preseason and postseason cognitive performance tasks such as the King-Devick test (KDT).

Hypothesis: Higher helmet forces recorded throughout a 10-week, 10-game youth football season would correlate with slower performance on postseason KDT.

Study Design: Prospective cohort study.

Level of Evidence: Level 3.

Methods: A cohort of 58 youth football players (ages 9-13 years) underwent pre- and postseason KDT. Players wore SpeedFlex helmets (Riddell) fitted with InSite Impact Response System helmet accelerometers (Riddell) which recorded impacts of ≥15*g*. Head impacts were tallied over a season and assigned a score of 1, 2, or 3 based on magnitude of *g* forces. Suspected concussions were correlated with KDT times and recorded instances of head impact. Pre- and postseason KDT scores were compared.

Results: During the season, 2013 head impacts were recorded. Median (range) total cumulative force score was 24 (5-476); 6 players sustained head impacts concerning for concussion, and 4 were clinically diagnosed with concussions. Overall, postseason KDT times improved compared with preseason, with a median (range) change of −4.8 seconds (−7.6, −1.1). Analysis showed no correlation between changes in KDT time and total cumulative force score over the season.

Conclusion: KDT times in youth football players did not change significantly based on head impact exposure over a single youth football season; most players' KDT times improved from preseason to postseason.

Clinical Relevance: Although our study did not show significant cognitive impact as measured by KDT over a single youth football season, the long-term effects of concussion on the immature brain and how it can impact cognitive development remains largely unknown and should be an area of ongoing study.

Keywords: concussion; concussion test; helmet accelerometer; sports-related concussion, youth athlete

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Approximately 2.8 million children between the ages of 7 and 14 years participate in youth tackle football in the United States.[38] Increasing media and public awareness of sports-related concussion (SRC) has been accompanied by an increase in knowledge about the potential cumulative effects caused by concussive and subconcussive impacts.[10] The 2022 international consensus statement on concussion in sport defines SRC as a "traumatic brain injury caused by a direct blow to the head, neck, or body resulting in an impulsive force being transmitted to the brain. . .initiating neurotransmitter and metabolic cascade, with possible axonal injury, blood flow change, and inflammation affecting the brain."[37] It has been reported that youth football players may not incur as many meaningful impacts to the head because they have smaller mass, a lower impact velocity, and participate less in high-risk contact sporting activities compared with adults.[29,47] However, the risks of concussions and prolonged recovery are higher in younger players.[11,13-15,26,36,47]

As research in the field of SRC has grown, one area of exploration has focused on impaired oculomotor function after trauma to the brain.[24] Studies have shown that vision-based assessment tools such as the King-Devick test (KDT) are more sensitive for detecting concussion compared with cognitive and balance testing alone.[20] The KDT is a quick vision-based sideline screening tool that provides an objective measure for determining concussion likelihood, and can be administered by medical or nonmedical personnel.[46] It is a timed test (recorded in seconds) assessing horizontal saccadic eye movements by reading lines of single-digit numbers that progressively become closer together.[31,41] Robust validation data from KDT indicate that, when compared with a baseline test, any slowing of test time or uncorrected errors increases the likelihood that an athlete has sustained a concussion by 5-fold.[19,30] While KDT is not an established marker of cumulative head trauma, studies have shown associations between repetitive head impacts and visual/vestibular impairments, including slower KDT in collegiate football players with repetitive head impacts, but no clinically significant symptoms.[19] It is unknown whether the same phenomenon will be observed in younger football players, thus the need for studies such as this. Repetitive head impacts have been associated with general neurocognitive slowing acutely, despite no major changes over the course of a season.[34] The same study discusses neuroimaging that showed decreased activation levels in areas of the prefrontal cortex and cerebellum. The lack of studies in youth football pertaining to the effects of cumulative head trauma led to this study.

The use of accelerometers worn in helmets of collegiate and professional American football players to analyze head impact injury biomechanics has increased knowledge of the velocities, direction, forces, accelerations, and frequency of head impacts.[28,47] On its own, head impact force measurement is not a sufficient evaluation tool for concussion due to its high false positive rate and lack of a standard force threshold for concussion.[36] However, head impact detection systems may prove to be a useful tool to identify head impact exposures (HIE) by measuring total subconcussive impacts per game and season.[28] Based on the Consensus Head Acceleration Measurement Practices (CHAMP) 2022 reporting guidelines, best practices for reporting studies using head kinematic measurements should include information about sensor technology and specifications, as well as analytical methods and data reporting. Ideally, improving the quality and consistency of research in this area will allow us to evaluate HIE more thoroughly.[3]

Despite youth American football players making up 70% of all American football participants, research has focused primarily on the professional and collegiate levels, and there is a paucity of concussion research done at the youth level.[7] The use of the KDT in conjunction with helmet accelerometers in youth players has not been documented extensively. Currently, only 2 studies have used KDT and rotational forces as variables but did not evaluate a correlation.[26,28] To our knowledge, no study has examined head impact forces and acceleration alongside changes in KDT scores across a youth football season.

The primary aim of this study was to evaluate whether higher helmet forces recorded throughout a 10-week, 10-game youth football season via InSite Impact Response System (Riddell) accelerometer technology would correlate with slower (poorer) performance on postseason KDT. Given the findings stated earlier in collegiate athletes, we hypothesized that repetitive head impacts over a single youth football season would lengthen the time for completion of the KDT. The secondary aim was to investigate the force magnitude, quantity, and timing of head impacts and to determine whether higher helmet impacts recorded via InSite Impact Response System would correspond with concussion symptomatology via point-of-care sideline KDT.

METHODS

Study Participants

The study protocol was reviewed and approved by the Mayo Clinic Institutional Review Board. A total of 68 youth American football players from the local Junior Development League (JDL) were recruited for study participation at their equipment pick-up day on July 20, 2019. Youth athletes signed an assent for participation, and parents or guardians signed consent. At equipment pick-up, all athletes underwent an initial preseason tablet-based KDT to electronically record baseline data. Per KDT protocol, athletes repeated the baseline KDT twice and the faster of the 2 scores was recorded as their baseline. Study participants were issued Riddell SpeedFlex helmets outfitted with InSite Impact Response System (Figure 1). The helmets have a battery life of 2 years, and the battery is exchanged during helmet reconditioning. The Riddell Sideline Monitor (RSM) detects the helmet signal within 50 yards and downloads the information to the RSM, which is subsequently uploaded to an online database. The information is stored on the helmets continuously until it is downloaded by the RSM. HIEs of ≥15*g* were recorded at practices (2 per week) and games (1 per

Figure 1. Representation of (a) Riddell Speed Flex Helmet and (b) Riddell InSite Impact Response System. Used with permission of Riddell, Inc., all rights reserved.

week) in a 10-week season. Of note, there were 3 weeks of practice during preseason with a total of 26 scheduled practices. After the final game on November 17, 2019, all study participants underwent a postseason KDT. If any players sustained a significant head impact during the season and were suspected of having a concussion, they were held out for the remainder of the practice or game. The injured players were evaluated on the sideline by the team physician who was fellowship trained and had a Certificate of Added Qualification (CAQ) in Primary Care Sports Medicine. The athletes underwent a postinjury KDT and were evaluated clinically based on the SCAT5 (Sport Concussion Assessment Tool 5th ed),[18] which includes a subjective symptom scale and physical examination findings including neurological, vestibular, balance assessment. The time of the recorded HIE was correlated to assess mechanism of injury via sideline witness and/or game video if available.

Statistical Analysis

Riddell's InSite Impact Response System records forces with a 15*g* threshold. In addition, the InSite system monitors forces received in manufacturer predetermined ranges of low (15-28*g*), medium (29-63*g*), and high (>63*g*) forces received. HIE was recorded for each player at each event (ie, practice or game) and a surrogate value of 1, 2, and 3 was assigned to low, medium, and high force, respectively, to calculate the cumulative force score for each event ($1N_{low} + 2N_{medium} + 3N_{high}$). Event scores for each player were added together after the final game to determine the total cumulative force received.

Statistical analysis of the primary aim included preseason and postseason KDT scores, as well as total cumulative force scores summarized as median and range. The correlation between postseason KDT scores and cumulative force was calculated using the Pearson correlation coefficient. For analysis of the secondary aim, medians and ranges of cumulative force were summarized for players with and without concussions, and the Wilcoxon rank sum test was used to evaluate the statistical significance of the difference as exploratory analysis.

RESULTS

Of 68 players initially recruited, 58 completed both pre- and postseason evaluation and were included in our study. Of 2013 impacts recorded, 1696 occurred during games, with the majority being low-impact forces (1424 low, 263 medium, and 9 high) and 317 during practices (267 low, 50 medium, and 0 high). Table 1 summarizes frequency of HIEs by helmet location and event, where most head impacts occurred on the rear side of the helmet and during games. The median (range) total cumulative force score was 24 (5-476).

Median baseline preseason KDT time was 54.7 seconds. Postseason median KDT time was 49.2 seconds, demonstrating a median improvement of −4.8 seconds from pre- to postseason. Players who received a total cumulative force score of ≤24 had a median improvement of −2.9 seconds, whereas those with cumulative force scores >24 had a median improvement of −5.4 seconds (Table 2). Our results suggest a moderately negative correlation ($r = -0.06$) between change of test duration and total HIE force (Figure 2).

During the 10-week season, 6 players sustained head impacts concerning for concussion that led to further evaluation by the sports medicine team physician on the sideline. Of these 6 players, 4 were clinically diagnosed with a concussion based on their symptoms and physical examination using the SCAT5. Per consensus guidelines,[37] some concussion symptoms can develop minutes to hours later; therefore, all 6 athletes were held from play the rest of the day. The 2 athletes not clinically diagnosed with a concussion had improved postinjury KDT times, and both had follow-up in clinic postinjury, where they remained asymptomatic and were deemed to not have a concussion. All 6 players who sustained concerning head impacts had postseason KDT times that improved by a median

Table 1. Summary of HIEs by location on helmet for 58 youth athletes

Location	Game	Practice	Total
Front	428	77	505
Back	478	118	596
Top	493	51	544
Right	153	24	177
Left	144	47	191
Total	1696	317	2013

HIE, head impact exposure.

Table 2. KDT duration by total cumulative force score

	≤24 (n = 29)	>24 (n = 29)	Total (N = 58)	*P* value
Preseason	52.9 (47.5, 64.5)	58.0 (49.2, 64.0)	54.7 (47.5, 64.4)	0.79
Postseason	48.8 (46.1, 57.0)	51.0 (44.6, 57.0)	49.2 (45.3, 57.0)	0.67
Change	−2.9 (−7.3, −1.0)	−5.4 (−9.3, −1.5)	−4.8 (−7.6, −1.1)	0.19

Variables are summarized in seconds as median (IQR). IQR, interquartile range; KDT, King-Devick test.

of −3.7 seconds compared with preseason (Table 3). Of the 4 players who were clinically diagnosed with a concussion, the majority sustained rear HIE and a median (range) total cumulative force score of 22 (10.0-45.2). Of the 4 athletes clinically diagnosed with concussion, 2 had longer postinjury KDT times, but improved postseason KDT times. The other 2 athletes diagnosed with concussion had shorter postinjury KDT times but longer postseason KDT times. The 2 athletes who sustained head impacts but were not clinically diagnosed with a concussion had improved KDT both postinjury and postseason (Figure 3). The 52 players who did not sustain head injuries improved their KDT times by a median of −4.8 seconds (Figure 3).

DISCUSSION

The effect of repetitive concussive and subconcussive head impacts on neurocognitive function continues to be a concern in contact sports, and there is a paucity of literature on this phenomenon in youth athletes. To our knowledge, this is the first study to correlate InSite Impact Response System readings with KDT scores in youth athletes (ages 9-13 years). During the 10-week season, 6 of 58 players sustained head injuries, 4 of whom were diagnosed clinically with concussion utilizing KDT and the SCAT5 clinical assessment. Postseason, most players had a faster KDT time, instead of the hypothesized slower time, compared with their preseason baseline KDT. And faster

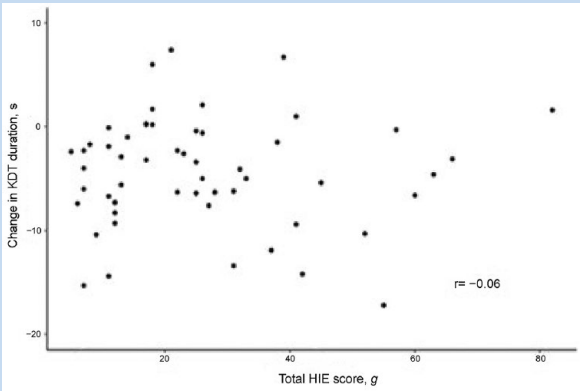

Figure 2. Change of KDT duration versus total HIE force. Scatterplot suggests a moderately negative correlation ($r = -0.06$) between change of KDT duration and total HIE force. HIE, head impact exposure; KDT, King-Devick test.

postseason test times were also seen in 2 of the 4 athletes with clinically diagnosed concussions (Table 3). There appeared to be no correlation between changes in KDT times and total cumulative force score over a single youth football season. In fact, results suggested a moderately negative correlation ($r = -0.06$) between change of test duration and total HIE force (Figure 2).

Table 3. KDT duration by presence or absence of suspected concussion

	No concussion (n = 52)	Concussion (n = 6)[a]	Total (N = 58)	*P* value
Preseason	55.8 (47.6, 64.7)	51.9 (47.8, 52.9)	54.7 (47.5, 64.4)	0.38
Postseason	50.2 (45.0, 57.0)	46.5 (46.0, 47.7)	49.2 (45.3, 57.0)	0.76
Change	−4.8 (−8.4, −1.4)	−3.7 (−5.8, 0.6)	−4.8 (−7.6, −1.1)	0.18

Variables are summarized in seconds as median (IQR). IQR, interquartile range; KDT, King-Devick test.
[a]Of the 6 players with suspected concussion who underwent postinjury KDT, only 4 were clinically diagnosed with concussion.

King-Devick Test

Several factors may explain why our study did not show a significant correlation between total cumulative force score and performance on KDT. First, repetitive HIEs have not been proven definitively to cause neurocognitive changes in this youth age group; therefore, we must acknowledge that a correlation simply may not exist, or that subtle changes may not be detected by KDT in this youth population. Another potential reason is the timing of the KDT. Preseason KDT was completed at equipment pick-up in July during summer break when children have been out of academics for 6 to 8 weeks. Postseason testing occurred in November when children had been back in school for nearly 4 months. A structured school environment contributes to continued brain development or neurocognitive growth,[22] resulting in enhanced reading ability and, consequently, may improve KDT time.[16,28] As KDT is based primarily on rapid number identification, it can capture many measures of brain function, including impaired rapid eye movement, attention, and reading skills. When reading text, the eyes move from word to word, alternating between patterns of fixations and saccadic movements, or rapid eye movements, between fixations.[40] These are some of the same skills that students develop throughout the school year with reading practice. These reading skills are incorporated into metrics that often determine school advancement and grade level placement. In fact, there is a phenomenon called the summer slide, describing the cumulative loss effect on reading skills that students are prone to over summer breaks, especially when they are younger (elementary or middle school age).[12] The effect of this summer learning loss may be evident in our preseason KDT results, since many of the same skills employed to read effectively in school are also employed in achieving faster KDT times.

Effort and repetition of the test may also play a role in the results, as it would be difficult to discern whether athletes were "trying" or "tanking" the test in either preseason or postseason testing. It may be advantageous in future studies to include a midseason KDT to minimize concern of the summer slide. In fact, it is recommended that children <10 years old retake a baseline test every 6 months, likely due to rapid increases in learning. Our study period was 4 to 5 months from equipment pick up in July to final game in November.

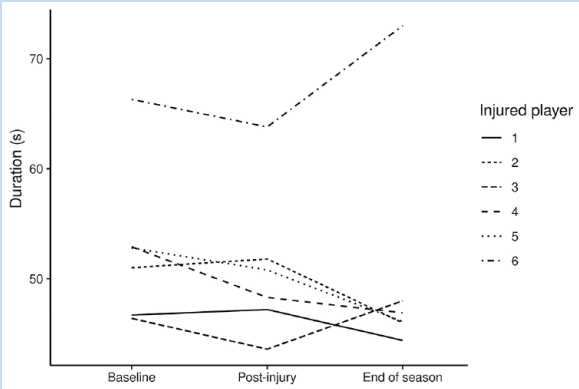

Figure 3. Change in KDT duration for players with suspected head injury. Players 1 and 2 were diagnosed with concussion and had longer duration (worsening KDT) postinjury and shorter duration (improved KDT) end of season (midseason injuries). Player 3 was diagnosed with concussion first game of season and had improved KDT postinjury compared with baseline and worsening KDT end of season (2.5 months later). Players 4 and 5 were not diagnosed with concussion and had improved KDT postinjury with further improvement end of season. Player 6 was diagnosed with concussion and had improved KDT postinjury (last practice of season) and worsening KDT compared with baseline at end of season (3 days later). KDT, King-Devick test.

Several studies have demonstrated improvement in KDT time with increased age.[2,21,28] In a study evaluating KDT time in a youth population (8-18 years), KDT time improved as age increased, with a plateau at age 14 years. There was a 31-second improvement corresponding to 75% change between ages 8 and 18 years.[21] As proposed by Galetta et al,[20] improvement in KDT time with increased age could be explained by ongoing brain development in youth athletes, as

seen on functional magnetic resonance imaging scans. The eye movement tasks of the KDT involve the frontal eye field circuits, in which continued frontal lobe change is seen throughout childhood.[32] In addition, improved saccadic eye performance in childhood may be explained by stronger activation of cortical eye fields.[1,5] These findings support potential limitations of baseline testing due to ongoing neurocognitive development, and thus support the recommendation to administer the KDT in youth athletes at least annually or even every sport season to maintain accurate baseline data.[28]

Brain development in the student athletes in our study from July to November may further explain the improved postseason KDT times. In addition, the inherent sharpening of skills during sports training may contribute to improved KDT times between the beginning and end of a season. For example, throughout the football season, players often improve reaction times, catching skills, and play reading through continuous practice and coaching. A small study by Munce et al[35] of 10 youth football players found that both occulomotor performance measured by KDT and reaction time assessed by the Immediate Postconcussion Assessment and Cognitive Test improved moderately to significantly throughout the course of a 12-week season. This study aligns with our findings and further supports the theory that gains in neurocognitive function during a season of football training could offset the negative neurocognitive effects of subconcussive impact being evaluated by the KDT. Future studies should be designed with the above reading development changes in mind, should similar goals of comparing KDT changes postcumulative HIE be a stated study objective.

The KDT is an excellent tool for identifying witnessed and unwitnessed concussions on the sideline with a sensitivity and specificity of 86% and 90%, respectively.[28] The reliability of KDT in detecting concussion is likely because visual processing requires a wide distribution of neuronal networks from cortical to subcortical and brainstem regions, capturing assessment of multiple domains, including attention, processing speed, motor planning, language, and visual-spatial integration.[20] In a study of 243 youth and 89 collegiate athletes, postinjury KDT scores were slower by an average of 5.2 seconds from baseline in athletes who sustained a concussion, and faster by an average of 6.4 seconds in nonconcussed controls after competition or practice.[20]

The KDT is a valuable tool in American football as contact sports are associated with a higher likelihood of head impacts, and thus increased risk of concussion and need for rapid sideline assessment.[6] As seen in our study, concussion remains a clinical diagnosis based on high suspicion from the athletes' symptoms and physical examination, such as the SCAT5 (most recent SCAT at the time of our study), and now the SCAT6.[17,18,23] The KDT is an additional tool a health provider can utilize easily to help with remove-from-play recommendations after a head injury,[27] especially when diagnostic uncertainty exists, but it must always be taken in context with the athlete's clinical presentation. As noted above, confidence of concussion diagnosis improves and bias decreases with use of multiple assessment tools. Therefore, in our study, using the KDT as a single neurocognitive test may have limited our findings.

Age Difference and HIE

The frequency and magnitude of head impacts in our youth study was quite low compared with a similar number of high school and collegiate football athletes. Schnebel et al[42] measured head impacts (>10*g*) of 40 collegiate level athletes and 16 high school athletes over all practices and games during a full football season. They recorded 54,154 impacts occurring over 93 practices and 12 games in the college group compared with 8326 impacts occurring over 15 practices and 9 games in the high school group. Overall, college players were found to sustain impacts >60*g* and >98*g* more frequently than high school players. In both groups, skill position players received more total impacts compared with linemen.[42] Unlike the volume of high impacts seen in high school and collegiate athletes, in our study, the total number of HIEs for the 58 athletes occurring over 26 practices and 10 games was much lower at 2013, with most being low-impact forces. Perhaps underestimating total HIE, a limitation of our study, was the preset threshold of the InSite accelerometers to record impacts only >15*g*. Studies have shown that almost 50% of impacts are removed with a 15*g* threshold compared with a 10*g* threshold.[28,36] In addition, we could not track the position of the player in relation to impacts, as the JDL structure rotates the youth athletes to different positions weekly.

Timing of HIE

In our current study, most head impacts occurred on the rear side of the helmet during games (Table 1) and were of low impact force (15-28*g*). The lower force and quantity of head impacts seen in our study compared with high school and college levels could be explained by several factors related to tackling techniques and JDL practice guidelines. First, timing of head impacts is an important concept both medically and for coaching purposes. Our study demonstrated that overall impacts occurred 5 times more in games (1696 HIE) versus practices (317 HIE) in JDL youth football. This is likely due to JDL following the USA Football National Practice Guidelines for Youth Football that limit live contact in practice to <30 minutes of full contact each practice, including "thud" and "live action" contact.[44] "Thud" contact consists of drills that are run at competitive speed with contact restricted to above the waist and players remain on their feet, whereas "live-action" contact incorporates drills completed in game-like conditions where players may be taken to the ground during contact. JDL also uses drills on air, bags, and control player versus player for other drills as defined by USA Football.[44] JDL practice plans are designed to maximize opportunities for all youth athletes to work on fundamentals of USA Football proper tackling and blocking instead of focusing on scheme and scrimmage.

Minimizing live contact is also practiced in other youth football leagues. In 2012, Pop Warner Football implemented

rule changes to minimize contact in an effort to reduce concussions in youth by prohibiting full-speed head-on blocking or tackling drills in which players are lined up >3 yards apart.[39] The new rules also limit the amount of contact allowed at each practice to a maximum of one-third of the practice time.[39]

A large cohort study investigated the effects of coaching education and contact prevention in practice after USA Football released Heads Up Football coaching education and Pop Warner contact restriction guidelines.[25] The study found decreased injury rates for groups that received both Heads Up Football coaching and practice contact restrictions compared with groups that did not. Of the 370 injuries reported, 64.3% occurred in the non-Heads Up Football group compared with 26.2% in the Heads Up Football group and 9.5% in the Heads Up Football and Pop Warner contact restriction group. Of the 45 concussions reported, 48.9% occurred in the non-Heads Up Football group compared with 35.6% in the Heads Up Football group and 15.6% in the Heads Up Football and Pop Warner group.[25]

Proper tackling technique has been shown to reduce head impacts.[25] One study involving high school football players investigated the effects of an intervention to incorporate targeted football drills focused on improved blocking and tackling techniques, with the goal of reducing exposure to subconcussive head impacts. After 1 month of training, frequency of head impacts greater than 15*g*, as measured by helmet accelerometer, demonstrated a significant reduction of approximately 30%, reinforcing the importance of proper coaching and tackling technique to reduce subconcussive head impacts.[9] Overall, our group of 58 JDL youth football players sustained only 4 clinically confirmed concussions over a 10-week season (6.9%). This low rate of injury may be due in part to the Heads-Up Football training of our coaches as well as JDL's limited contact approach. Further analysis of the 4 concussions in our study indicated that the mechanism of injury for all was rear helmet to turf. Given the USA Football training and instruction in shoulder tackling, these youth athletes were not as likely to have a helmet-to-helmet impact. More often, they get knocked down and may not have the neck strength to keep their head off the turf when they are going to the ground. This is further supported by the fact that the greatest number of overall impacts recorded by the helmet accelerometer were to the back of the head.

In addition, higher impact forces may be seen in specific positions and locations on the football field. Campolettano et al[7] demonstrated that the players in back positions who were more likely to play in the open field had higher rates of HIE compared with other positions in youth football. However, JDL football allows players to develop skills at each position, as they rotate positions weekly, which may have neutralized this repetitive exposure risk for dedicated position players. In essence, JDL-style training; with weekly variance of positions, limitations of overall contact, and USA football tackling techniques; may be protective from repetitive HIE and thus protective from concussion in youth athletes.

Helmet Accelerometers

Currently, helmet accelerometers are helpful for research, as in our study, and for coaching to monitor tackling techniques. However, it is understood that these accelerometers are indirect measures of force and are not used to diagnose concussion.[23] Current literature debates whether resultant linear acceleration, resultant rotational acceleration, or combined resultant linear and rotational acceleration influence concussion risk.[36] Currently, there is no criterion-standard threshold *g*-force head impact that reliably induces a concussion injury. In addition, it is impossible to directly measure impact to the brain itself via an accelerometer in the helmet.[37] Some athletes may develop concussion symptoms after low-force impacts, while high-force impacts may be sustained without injury. It is also difficult to ascertain whether the accumulation of subconcussive impacts leads to concussion symptomatology. Subconcussive injury is defined as forces transmitted to the brain that may cause axonal injury without clinical symptoms.[33] It is currently unknown if subconcussive blows cause subtle cognitive impairment over time.[45] However, helmet accelerometers may prove useful as measurement of total subconcussive impacts per game and season, as well as guide technique and contact time adjustments that lead to fewer impacts.[9]

Special Considerations for Youth Athletes

Although our data did not suggest subconcussive cognitive decline via KDT over a single youth football season, several theories exist as to why youth players may be vulnerable to injury from less forceful impacts, including (1) incomplete myelination, (2) reduced neck musculature size and strength, and (3) more variability in body build.[8,11,13] The frontal lobe, responsible for working memory, undergoes a growth spurt during adolescence, making it more fragile and vulnerable to concussions.[4] Long-term concussion effect on the immature brain and how it can impact development is still being studied, and the Concussion in Sport Group has proposed a multidisciplinary working group be established to guide research in this area.[26,37] Even in the absence of a clinically diagnosed concussion, there may be more subtle neurocognitive changes from these impacts due to microstructural injury and neurometabolic changes in the brain.[43] For this reason, concussion prevention and early identification is extremely important. Evaluation of these variables will help expedite identification of the concussed athlete, leading to modifications in contact technique, protective equipment, and rules aimed at mitigating concussion risk, especially in the youth athlete population.

Determining whether a specific quantity and force of subconcussive head impacts affects risk of concussion could potentially identify higher risk athletes. This information could also help create guidelines on head impact limitations analogous to baseball, where exceeding pitch count recommendations increases risk of upper extremity injury.

CONCLUSION

Overall, our study did not demonstrate a correlation between HIE and performance on KDT over a 10-week youth football season. Long-term concussion effect on the immature brain and how it can impact cognitive development remains largely unknown and should be an area of ongoing study. For this reason, concussion prevention via education on proper sporting technique, such as tackling, and early concussion identification using quick, easy-to-use, validated sideline tools, such as KDT, are extremely important. Coaches and sports medicine staff may use head impact technology like the InSite Impact Response System to evaluate the variables involved with concussive and subconcussive impacts, including location of impacts and force, but not to conclusively diagnose concussion. Evaluation of these variables may lead to modifications to style of play, including rule changes, use of proper protective gear, and improved tackling techniques aimed at mitigating concussion risk, especially in the youth athlete population.

Clinical Recommendations

The results of this study show no correlation between KDT results and HIE. However, it did support the usefulness of head impact technology like the InSite Impact Response System in evaluating variables involved with concussive and subconcussive impacts, including location of impacts and force. The Strength of Recommendation Taxonomy is C for this evidence, as more studies need to be done regarding the long-term effects of such forces on the immature brain and how it can impact cognitive development. Currently, head impact technology such as accelerometers, may not be used to conclusively diagnose concussion, although they may be useful for sports medicine staff and coaches to evaluate the location of head impacts and forces. Strength of Recommendation Taxonomy is B for this evidence as, clearly, more variables were at play that were needed to diagnose concussions, not just the nature of the forces transmitted to the heads of the young athletes included in the study.

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